

Applied Statistical Modeling 2

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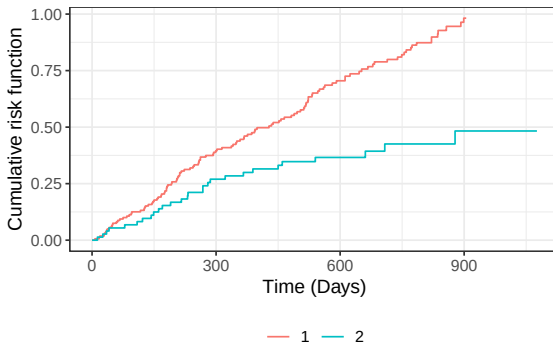
Institute of Management - Scuola Superiore Sant' Anna, Pisa

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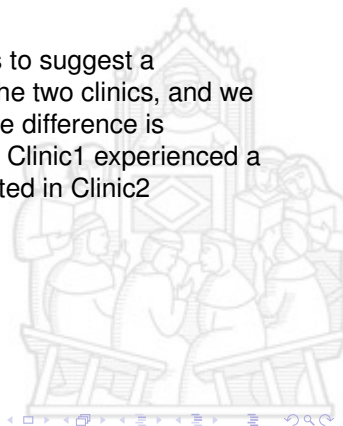
Cox proportional hazard model

Consider the example of the *addicts* dataset and go on with the comparison of survival experience in the two clinics



Cox proportional hazard model

The plot of the cumulative risk functions seems to suggest a difference in the survival experience between the two clinics, and we already verified using the Log-Rank test that the difference is statistically significant, thus, subjects treated in Clinic1 experienced a higher risk of failure as compared to those treated in Clinic2



Cox proportional hazard model

Let's try to estimate the effect of the variable *Clinic* using the Cox model, by typing in R

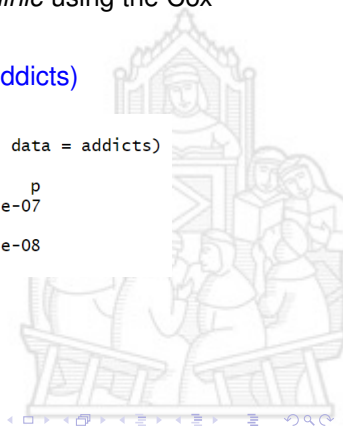
```
m1<-coxph(Surv(survt, status) ~ clinic, data = addicts)
```

Call:

```
coxph(formula = Surv(survt, status) ~ clinic, data = addicts)
```

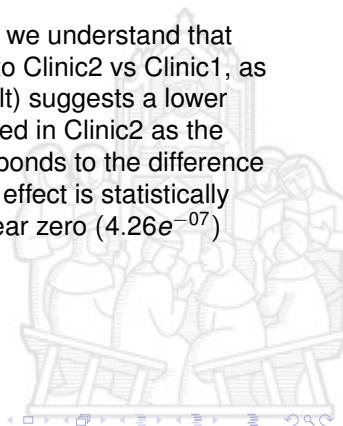
	coef	exp(coef)	se(coef)	z	p
clinic	-1.0754	0.3412	0.2127	-5.057	4.26e-07

Likelihood ratio test=30.99 on 1 df, p=2.594e-08
n= 238, number of events= 150



Cox proportional hazard model

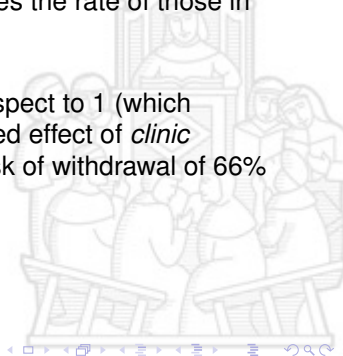
Just focusing for the moment on the estimates, we understand that the estimated coefficient, $\hat{\beta}$, (which is referred to Clinic2 vs Clinic1, as R takes the lowest value as reference by default) suggests a lower risk of withdrawn from treatment for those treated in Clinic2 as the estimate is negative, $\hat{\beta} = -1.075$ which corresponds to the difference in the log hazard for Clinic2 versus Clinic1, the effect is statistically significant as the p-value for the Wald test is near zero ($4.26e^{-07}$)



Cox proportional hazard model

Using the exponentiated coefficients which could be interpreted in terms of hazard ratio, we have $\exp(\hat{\beta}) = \hat{HR} = 0.341$, suggesting that subjects in Clinic2 withdraw at about 0.341 times the rate of those in Clinic1

Being a ratio, we can also interpret $\hat{HR}$ with respect to 1 (which indicates equal rates) and say that the estimated effect of *clinic* translates into an estimated reduction of the risk of withdrawal of 66% in Clinic2 vs Clinic1 over the study period



Cox proportional hazard model

The 95% confidence interval for $\hat{HR}$ could be obtained typing

`exp(confint(m1))`

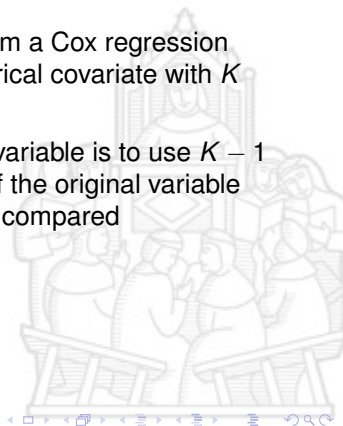
which provides a 95% with limits going from 0.225 to .518



Cox proportional hazard model

To broaden the interpretation of coefficients from a Cox regression model, let's consider now the case of a categorical covariate with K distinct values

A commonly used approach to deal with such variable is to use $K - 1$ dummy variables choosing one of the k level of the original variable as reference, against which all other levels are compared



Cox proportional hazard model

Using this approach we can estimate $k - 1$

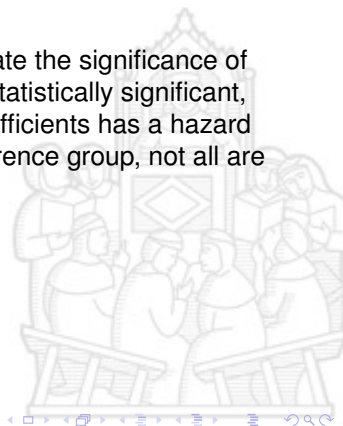
$$\hat{HR}_k = e^{\hat{\beta}_k}$$

each estimating the hazard ratio for the k^{th} group to that of the reference group



Cox proportional hazard model

Under this setting the G statistic used to evaluate the significance of the overall model coefficients, when resulting statistically significant, suggests that at least one of the estimated coefficients has a hazard rate that is statistically significant from the reference group, not all are significant (!)

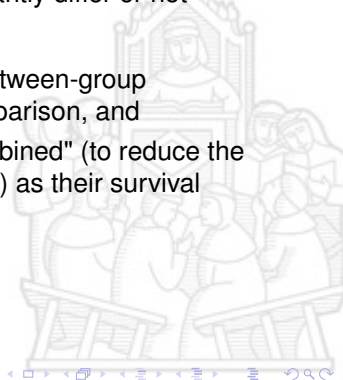


Cox proportional hazard model

When using a categorical variable, we may also be interested in understanding whether two of the $\hat{H}\hat{R}_k$ significantly differ or not

This could be of interest

- ▶ to answer a specific question related to between-group comparison, in addition to the overall comparison, and
- ▶ to understand if two groups could be "combined" (to reduce the number of possible values of the covariate) as their survival experience does not differ



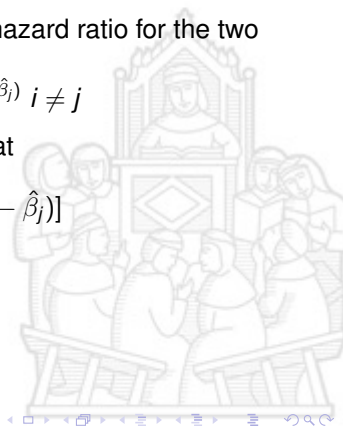
Cox proportional hazard model

This could be done using the estimator of the hazard ratio for the two groups

$$\hat{HR}(k = i \text{ versus } k = j) = e^{(\hat{\beta}_i - \hat{\beta}_j)} \quad i \neq j$$

and its 95% confidence interval considering that

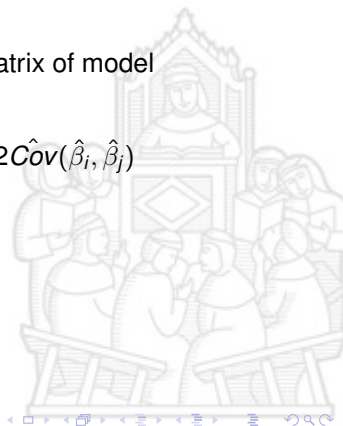
$$\exp[(\hat{\beta}_i - \hat{\beta}_j) \pm 1.96 \times \hat{SE}(\hat{\beta}_i - \hat{\beta}_j)]$$



Cox proportional hazard model

Calculating $\hat{SE}(\hat{\beta}_i - \hat{\beta}_j)$ from the covariance matrix of model coefficients considering that

$$\hat{Var}(\hat{\beta}_i - \hat{\beta}_j) = \hat{Var}(\hat{\beta}_i) + \hat{Var}(\hat{\beta}_j) - 2\hat{Cov}(\hat{\beta}_i, \hat{\beta}_j)$$



Cox proportional hazard model

The Wald test for the difference $\hat{\beta}_i - \hat{\beta}_j$ is

$$z = \frac{\hat{\beta}_i - \hat{\beta}_j}{\hat{SE}(\hat{\beta}_i - \hat{\beta}_j)}$$

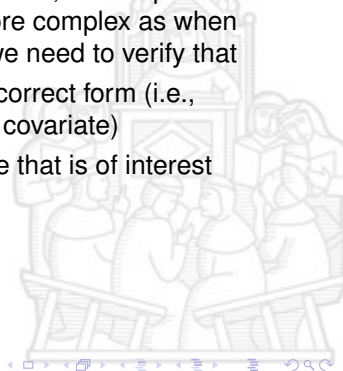
which follows the standard normal distribution



Cox proportional hazard model

While the interpretation of model coefficient in a Cox model using a single continuous covariate could be thought easier, as compared to dummy or categorical variables, it is indeed more complex as when handling continuous covariate in a Cox model, we need to verify that

- ▶ the variable is included in the model in its correct form (i.e., assume that the log hazard is linear in the covariate)
- ▶ the meaningful unit change in the covariate that is of interest (i.e., 1-unit change, 5-unit change)



Cox proportional hazard model

The inclusion of the variable in the model in its correct form needs to be properly verified, i.e., assessing the linear assumption



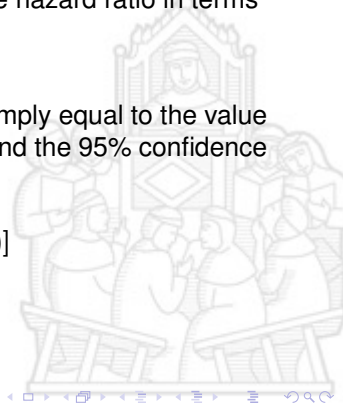
Cox proportional hazard model

While, if the meaningful unit change is in terms of cx (where c is a constant), we need to take into account that the hazard ratio in terms of cx is given by

$$\hat{HR}(c) = e^{c\hat{\beta}}$$

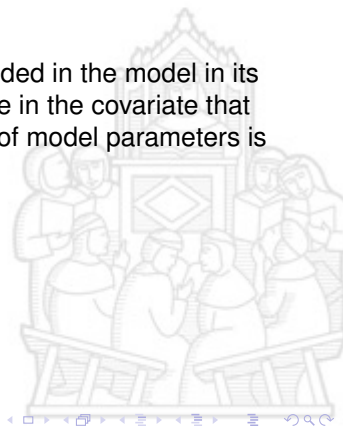
so the change in terms of cx rather than c is simply equal to the value of the change of interest times the coefficient and the 95% confidence interval could be obtained using

$$\exp[c\hat{\beta} \pm z_{1-\alpha/2}|c|\hat{SE}(\hat{\beta})]$$



Cox proportional hazard model

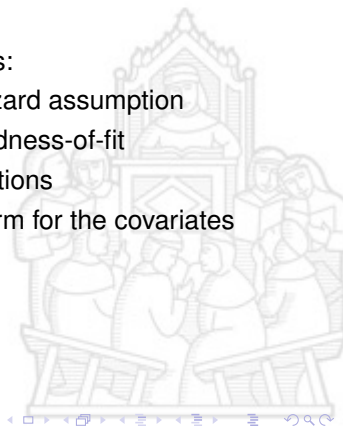
Only once we have verified the variable is included in the model in its correct form and we the meaningful unit change in the covariate that is of interest have been defined, interpretation of model parameters is straightforward



Assessing adequacy of a Cox proportional hazard model

Evaluating adequacy of the Cox model involves:

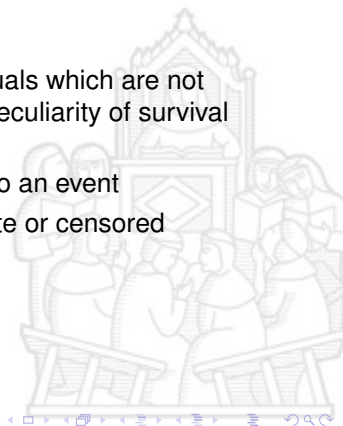
- ▶ assessing and testing the proportional hazard assumption
- ▶ computing summary measures of the goodness-of-fit
- ▶ assessing leverage and influence observations
- ▶ assessing the correctness of the model form for the covariates (i.e., linearity assumption)



Assessing adequacy of a Cox proportional hazard model

All these evaluation could be done using residuals which are not straightforward to be defined considering the peculiarity of survival analysis, because

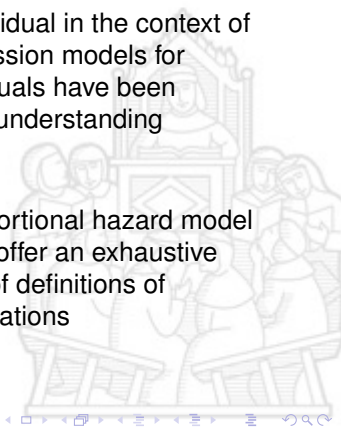
- ▶ the dependent variable of interest is time to an event
- ▶ the observed values are typically incomplete or censored



Assessing adequacy of a Cox proportional hazard model

As there is no an obvious way to determine residual in the context of a Cox proportional hazard model (and in regression models for survival data in general), several different residuals have been proposed and each serves to a different goal in understanding adequacy of a fitted proportional hazard model

Some examples of residuals defined for a proportional hazard model are given in the next slides without claiming to offer an exhaustive overview, but just to illustrate some examples of definitions of residuals that may be useful for practical applications



Assessing adequacy of a Cox proportional hazard model

Schoenfeld and scaled Schoenfeld residuals are generally used to assess the proportional hazard assumption

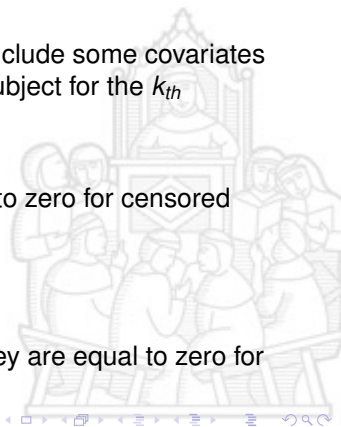
Having observed a set of survival data which include some covariates x_i , the Schoenfeld residuals for the generic i^{th} subject for the k_{th} covariate is

$$\hat{r}_{ik} = c_i(x_{ik} - \hat{x}_{w,k})$$

where c_i indicating censoring and being equal to zero for censored subjects, and

$$\hat{x}_{w,k} = \frac{\sum x_{jk} e^{x'_j \hat{\beta}}}{\sum e^{x'_j \hat{\beta}}}$$

the sum of Schoenfeld residuals is zero and they are equal to zero for all censored subjects

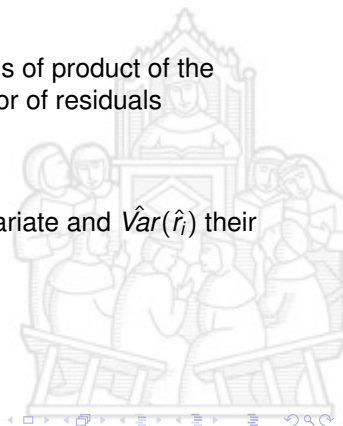


Assessing adequacy of a Cox proportional hazard model

Scaled Schoenfeld residuals are defined in terms of product of the inverse of the covariance matrix times the vector of residuals

$$\hat{r}_i^* = [\hat{Var}(\hat{r}_i)]^{-1} \hat{r}_i$$

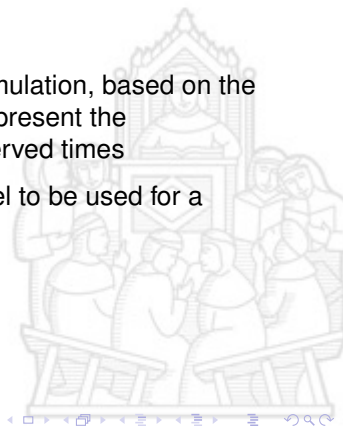
where $\hat{r}_i$ is the vector of residuals for the k covariate and $\hat{Var}(\hat{r}_i)$ their covariance matrix



Assessing adequacy of a Cox proportional hazard model

Martingale residuals have a more complex formulation, based on the possibility of having a counting process that represent the proportional hazard model at the different observed times

They are useful to understand the proper model to be used for a covariate included in the model



Developing a Cox proportional hazard model

In practical application, we are generally interested in including multiple covariates in a Cox model, this is done not only to assess the effect of multiple covariates of interest, but also in the case we are interested in a single covariate but we need to "control for confounding"



Developing a Cox proportional hazard model

Indeed, when using multiple variable in a model (either Cox or whatever regression model), different situations may occur for a covariate x_j that could results

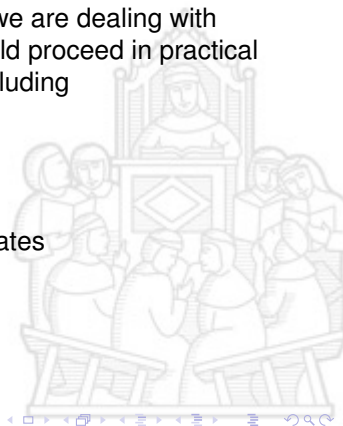
- ▶ a confounder of the relation between the main covariate of interest x_i and the outcome of interest
- ▶ an effect modifier of the relation
- ▶ nor a confounder, nor an effect modifier



Developing a Cox proportional hazard model

Broadly speaking, to understand the situation we are dealing with with respect to the covariates x_i and x_j , we could proceed in practical terms by fitting different regression models, including

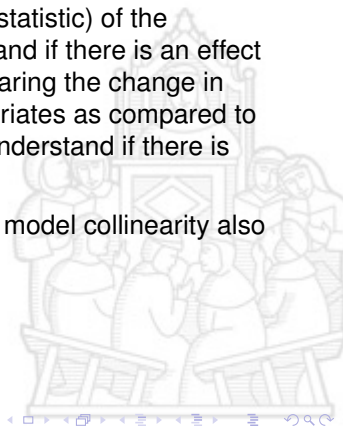
- ▶ x_i but not x_j as covariate
- ▶ x_j but not x_i as covariate
- ▶ both x_i and x_j as covariates
- ▶ x_i, x_j and their interaction $x_i \times x_j$ as covariates



Developing a Cox proportional hazard model

By assessing the significance (using the Wald statistic) of the interaction term $x_i \times x_j$ we are able to understand if there is an effect modification, if this could be excluded by comparing the change in model coefficient in models containing both covariates as compared to the one containing just one covariate we can understand if there is confounding

When more than one covariate is included in a model collinearity also needs to be assessed

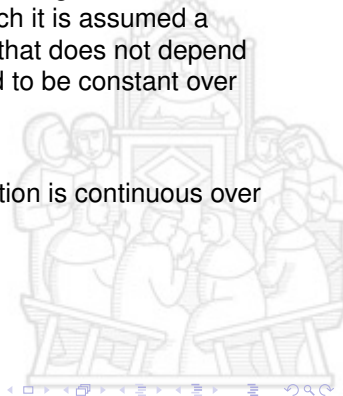


Extension of the Cox proportional hazard model

The Cox proportional hazard model allows modelling survival data efficiently and easily, specifying a model in which it is assumed a common unspecified baseline hazard function that does not depend on time and the effect of covariates is assumed to be constant over time

$$h(t, x, \beta) = h_0(t)e^{x\beta}$$

The Cox model also assumes that the observation is continuous over time and that subjects could be right censored

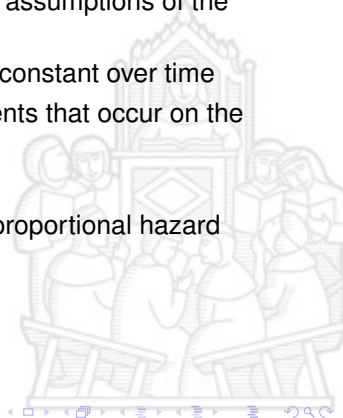


Extension of Cox proportional hazard model

However, there are circumstances in which the assumptions of the Cox model do not hold because

- ▶ the effect of one or more covariates is not constant over time
- ▶ there are competing events or multiple events that occur on the same subject
- ▶ data are interval censored

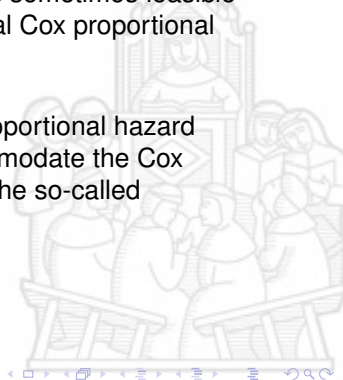
in all these cases, we need to extend the Cox proportional hazard model or even use different regression models



Extension of Cox proportional hazard model

Extending the Cox proportional hazard model is generally easier than using alternative regression models, and this is sometimes feasible when the assumptions underlying the traditional Cox proportional hazard model are not verified

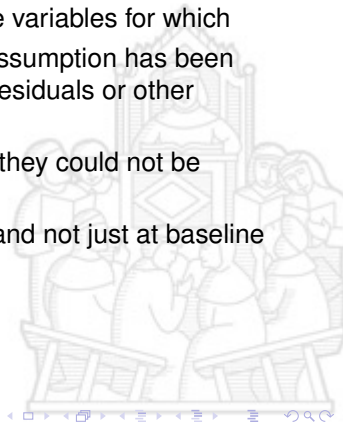
In particular, when some deviation from the proportional hazard assumption is detected, it is possible to accommodate the Cox proportional hazard model for the inclusion of the so-called *time-dependent* or *time-varying* covariates



Extension of Cox proportional hazard model

Time-dependent or time-varying covariates are variables for which

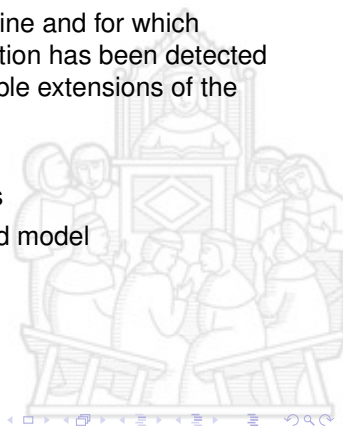
- ▶ a deviation from the proportional hazard assumption has been detected by the inspection of Schoenfeld residuals or other approaches discussed
- ▶ it is well known from previous studies that they could not be considered constant over time
- ▶ data are collected continuously over time and not just at baseline



Extension of Cox proportional hazard model

In the case of covariates collected just at baseline and for which deviation from the proportional hazard assumption has been detected or it is well-known from previous studies, possible extensions of the Cox proportional hazard model include

- ▶ the inclusion of *time-dependent* covariates
- ▶ the use of the stratified proportional hazard model



"Cox proportional hazard model" with time-varying covariate

When a covariate measured just at baseline does not satisfy the proportional hazard assumption, we have already discussed the possibility of eventually including it in a proportional hazard model by modelling its interaction with a function (i.e., linear or whatever is appropriate)

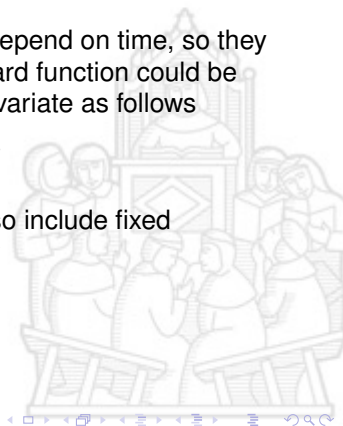
However, both in this case and in the case of variables that are recorded on multiple occasions over time (or could be derived, for example, age that could be derived by adding observation time to baseline age), it is possible to "appropriately" extend the Cox proportional model to account for time-varying covariate

"Cox proportional hazard model" with time-varying covariate

Indicating with $x(t)$ a covariate whose values depend on time, so they are determined by time t , the proportional hazard function could be generalized by the inclusion of time-varying covariate as follows

$$h(t, x(t), \beta) = h_0(t) e^{x(t)\beta}$$

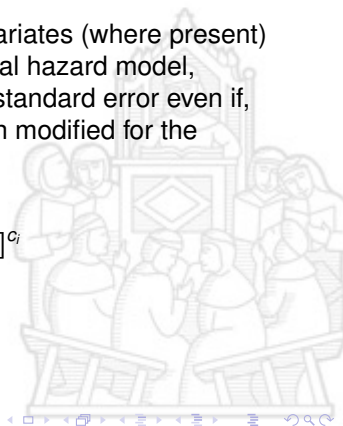
which is no longer proportional, even it may also include fixed covariates, those for which $x(t) = x(t = 0) = x$



"Cox proportional hazard model" with time-varying covariate

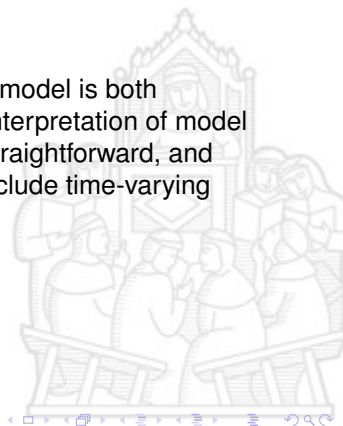
In such a model, the interpretation of fixed covariates (where present) is the same as in the traditional Cox proportional hazard model, similar is also the derivation of the associated standard error even if, obviously, the partial likelihood function is again modified for the inclusion of time-varying covariates as follows

$$l_p(\beta) = \prod_{i=1}^n \left[\frac{e^{x_i(t_i)\beta}}{\sum_{l \in R(t_i)} e^{x_l(t_i)\beta}} \right]^{c_i}$$



"Cox proportional hazard model" with time-varying covariate

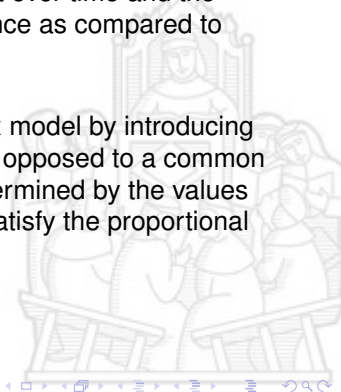
The inclusion of time-covariate in a regression model is both conceptually and practically hard, indeed the interpretation of model coefficients for time-varying covariates is not straightforward, and different approaches could be considered to include time-varying covariates in a regression model



Stratified Cox proportional hazard model

Stratified Cox proportional hazard model could be used when the effect of one (or more) covariate is not constant over time and the estimation of its effect is of secondary importance as compared to other covariates

The stratified model extends the traditional Cox model by introducing a stratum-specific baseline hazard function (as opposed to a common baseline hazard function) where strata are determined by the values of the covariates showing or known to do not satisfy the proportional hazard assumption

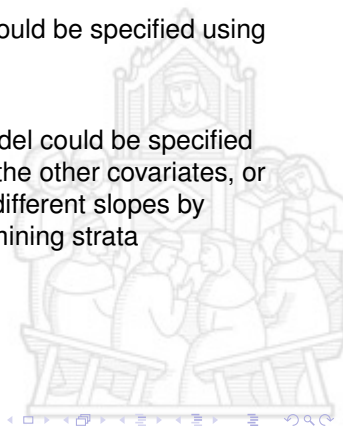


Stratified Cox proportional hazard model

The stratified Cox proportional hazard model could be specified using the following

$$h_s(t, x, \beta) = h_{s0}(t)e^{x\beta}$$

where $s = 1, 2, \dots, S$ are the strata, and the model could be specified assuming a common slope (over strata) for all the other covariates, or even accommodating the possibility of having different slopes by adding the interaction with the covariate determining strata



Stratified Cox proportional hazard model

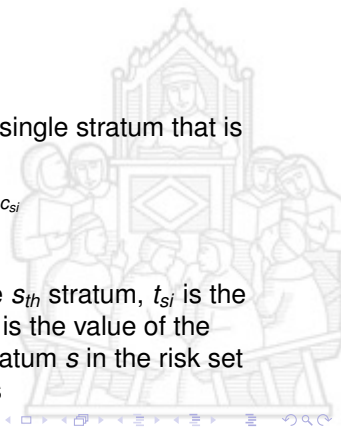
Even if the partial log-likelihood has a different form, as it is derived considering the product of the contributions of each stratum

$$l_{sp}(\beta) = \prod_{i=1}^S l_{sp}(\beta)$$

where $l_{sp}(\beta)$ represents the contribution of the single stratum that is

$$l_{sp}(\beta) = \prod_{i=1}^{n_s} \left[\frac{e^{x_{si}\beta}}{\sum_{j \in R(t_{si})} e^{x_{sj}\beta}} \right]^{c_{si}}$$

with n_s being the number of observations in the s_{th} stratum, t_{si} is the i_{th} observed value of time in the s_{th} stratum, c_{si} is the value of the censoring status at t_{si} , $R(t_{si})$ are subjects in stratum s in the risk set at time t_{si} , while x_{si} is the vector of p covariates



Stratified Cox proportional hazard model

Interpretation of model coefficients, their significance, and all the other approaches discussed to assess model assumptions remain the same as discussed for the traditional Cox proportional hazard model

Only in the case of a model specified allowing the possibility to have different slopes for the different strata, there is the need to pay attention to the interpretation of model coefficients taking into account the interaction between the covariate of interest and the covariate identifying strata

Stratified Cox proportional hazard model

Consider the proportional hazard Cox model adapted to the *addicts* data and aiming at assessing the effect of *methadone dose* and *clinic* on treatment withdrawal

```
coxph(formula = surv(survtime, status) ~ dose + clinic, data = addicts)
```

```
n= 238, number of events= 150
```

	coef	exp(coef)	se(coef)	z	Pr(> z)	
dose	-0.034327	0.966255	0.006274	-5.471	4.47e-08	***
clinic	-0.950878	0.386402	0.212079	-4.484	7.34e-06	***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
dose	0.9663	1.035	0.9544	0.9782
clinic	0.3864	2.588	0.2550	0.5855

```
Concordance= 0.659 (se = 0.025 )
```

```
Likelihood ratio test= 60.79 on 2 df, p=6e-14
```

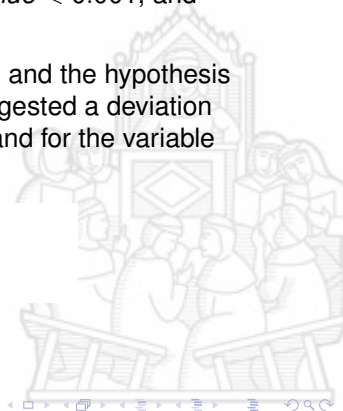
Stratified Cox proportional hazard model

Results suggest a statistically significant effect

- ▶ for both the dose, with $HR = 0.965$ $P - value < 0.001$, and
- ▶ clinic, $HR = 0.386$ $P - value < 0.001$

Anyway, the inspection of Schoenfeld residuals and the hypothesis testing for proportional hazard assumption suggested a deviation from the PH assumption for the overall model and for the variable *clinic*

	chisq	df	p
dose	0.373	1	0.5412
clinic	9.612	1	0.0019
GLOBAL	10.542	2	0.0051



Stratified Cox proportional hazard model

If the effect of the variable *clinic* is of secondary importance, as compared to *methadone dose*, it is possible to adapt a Cox proportional hazard model including *methadone dose* as covariate and the variable *clinic* as strata

```
Call:
coxph(formula = Surv(survt, status) ~ dose + strata(clinic),
      data = addicts)

n= 238, number of events= 150

            coef exp(coef)  se(coef)      z Pr(>|z|)
dose -0.033752  0.966812  0.006327 -5.334 9.59e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

            exp(coef) exp(-coef) lower .95 upper .95
dose            0.9668         1.034    0.9549    0.9789

Concordance= 0.634 (se = 0.025 )
Likelihood ratio test= 28.66 on 1 df,  p=9e-08
Wald test              = 28.45 on 1 df,  p=1e-07
```

Stratified Cox proportional hazard model

The HR for dose is slightly different from that one obtained in the traditional proportional hazard model, but is still statistically significant

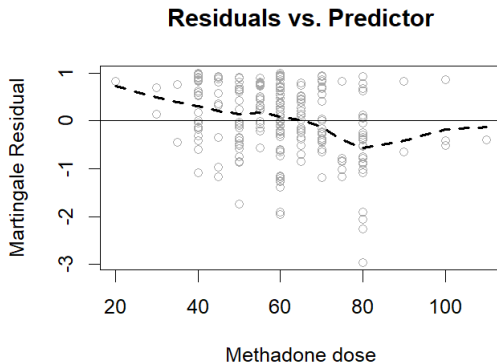
Moreover, stratifying there is no problem of deviation from the proportional hazard assumption for the overall model

	chisq	df	p
dose	0.904	1	0.34
GLOBAL	0.904	1	0.34



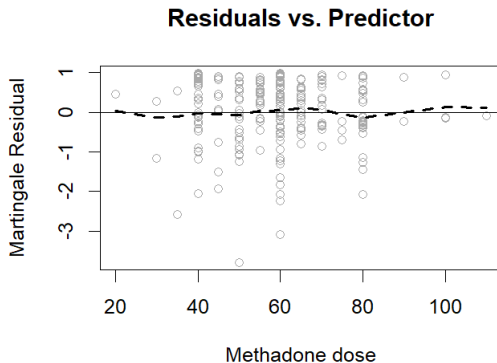
Stratified Cox proportional hazard model

In addition, while in the traditional model the plot of Martingale residuals showed a deviation from linearity of *methadone dose*



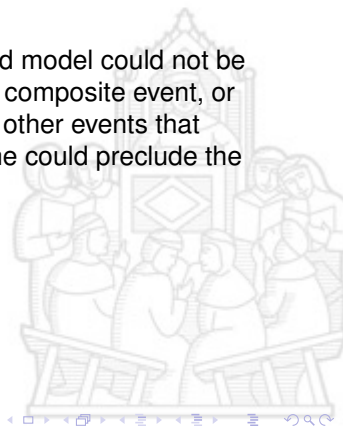
Stratified Cox proportional hazard model

With the stratified model also the linearity assumption seems to be met for *methadone dose*



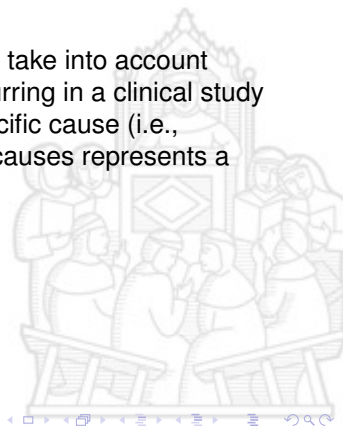
Competing risk models

A situation in which the Cox proportional hazard model could not be used occurs when the event of interest is not a composite event, or even it is a specific type of event and there are other events that "compete" with the event of interest because they could preclude the possibility of experiencing the event of interest



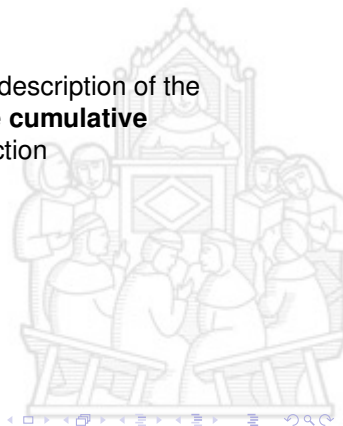
Competing risk models

The typical situation where there is the need to take into account competing events is, for example, the one occurring in a clinical study where the event of interest is death from a specific cause (i.e., cardiovascular disease) and death from other causes represents a competing event



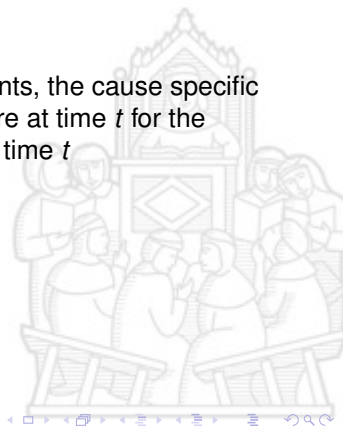
Competing risk models

When competing risks are present, the proper description of the survival experience is obtained considering the **cumulative incidence function (CIF)**, not the survival function



Competing risk models

Assume dealing with $j = 1, 2, \dots, k$ types of events, the cause specific hazard function $h_j(t)$ describes the rate of failure at time t for the event j , given that subjects survived till at least time t



Competing risk models

If the k types of events are mutually exclusive, the overall hazard function could be obtained as

$$h(t) = \sum_{j=1}^k h_j(t)$$

thus the overall event-free survival function is

$$S(t) = e^{-H(t)}$$

where $H(t) = \sum_{j=1}^k H_j(t)$ is the overall cumulative hazard function and $H_j(t)$ the cause specific cumulative hazard

$$H_j(t) = \int_0^t h_j(u) du$$



Competing risk models

The event-specific distribution function, called *subdistribution* or *cumulative incidence function (CIF)*, is then

$$F_j(t) = P(T \leq t, C = j) = \int_0^t h_j(u) S(u) du$$

and the overall distribution function is

$$F(t) = \sum_{j=1}^k F_j(t)$$

and $S(t) = 1 - F(t)$



Competing risk models

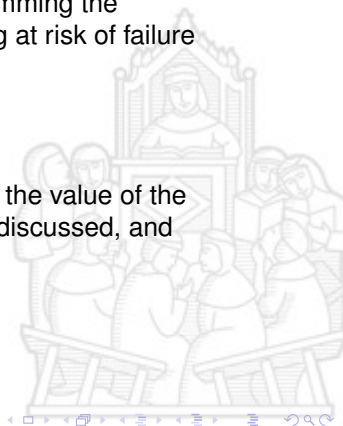
Suppose having a sample of n observation of follow-up time and event type (t_i, c_i) , the CIF is the cumulative proportion of failure due to cause j at time t and could be estimated by summing the cause-specific failure rates conditional on being at risk of failure

$$\hat{F}_j(t) = \sum_{t_{(i)} \leq t} \hat{h}_j(t) \hat{S}(t_{i-1})$$

where t_i is the largest event time and $\hat{S}(t_{i-1})$ is the value of the Kaplan-Meier estimator that has already been discussed, and

$$\hat{h}_j(t) = \frac{d_{ij}}{n_i}$$

is the event-specific hazard estimator



Competing risk models

Consider a simple example of ten subjects for which we recorded the occurrence of two different competing events and related follow-up time

Table: Hypothetical data

Id	Time	Event type
1	9	2
2	27	0
3	18	0
4	30	1
5	10	1
6	33	2
7	13	1
8	19	2
9	24	2
10	4	1



Competing risk models

The estimated cumulative survival and the cause-specific hazard could be obtained by considering the calculus below

Table: Kaplan-Meier estimate of the overall survival probability and cause-specific hazard for the hypothetical data

Time	Event type	$\hat{S}(t_i)$	n_i	d_{i1}	d_{i2}	$h_1(t_i)$	$h_2(t_i)$
0	-	1	10	0	0	0	0
4	1	0.9	10	1	0	0.1	0
9	2	0.8	9	0	1	0	0.111
10	1	0.7	8	1	0	0.125	0
13	1	0.6	7	1	0	0.143	0
18	0	0.6	6	0	0	0	0
19	2	0.48	5	0	1	0	0.2
24	2	0.36	4	0	1	0	0.25
27	0	0.36	3	0	0	0	0
30	1	0.18	2	1	0	0.5	0
33	2	0	1	0	1	0	1

Competing risk models

Then the cause-specific cumulative incidence function could be obtained as shown below

Time	$\hat{S}(t_j)$	$h_1(t_j)$	$h_2(t_j)$	l_{1t_j}	l_{2t_j}	CIF_1	CIF_2
0	1	0	0	0	0	0	0
4	0.9	0.1	0	0.1	0	0.1	0
9	0.8	0	0.1111	0	0.1	0.1	0.1
10	0.7	0.125	0	0.1	0	0.2	0.1
13	0.6	0.143	0	0.1	0	0.3	0.1
18	0.6	0	0	0	0	0.3	0.1
19	0.48	0	0.2	0	0.12	0.3	0.22
24	0.36	0	0.25	0	0.12	0.3	0.34
27	0.36	0	0	0	0	0.3	0.34
30	0.18	0.5	0	0.18	0	0.48	0.34
33	0	0	1	0	0.18	0.48	0.52

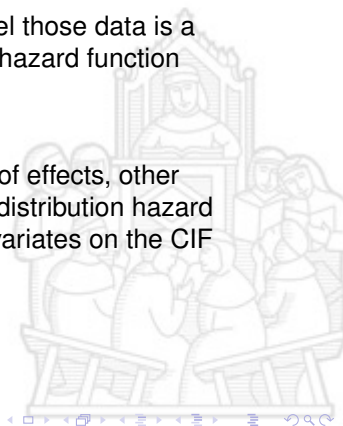
where $l_{ji} = h_j(t_i) * S(t - 1)$

Competing risk models

The simplest model that could be used to model those data is a proportional hazard model with cause-specific hazard function

$$h_{0j}(t)e^{x\beta}$$

which also allows for cause-specific estimates of effects, other common approach is to use the Fine-Gray subdistribution hazard model that allows for modeling the effect of covariates on the CIF



Competing risk models

Consider some examples of using the *bmtcrr* data

<i>Variable</i>	<i>Description</i>	<i>Statistical summary^a</i>
Sex	Sex	M = Male (100) F = Female (77)
D	Disease	ALL (73) AML (104)
Phase	Phase	CR1 (47) CR2 (45) CR3 (12)
Source	Type of transplant	Relapse (73) BM + PB (21) PB (156)
Age	Age of patient (years)	4–62 30.47 (13.04)
Ftime	Failure time (months)	0.13–131.77 20.28 (30.78)
Status	Status indicator	0 = censored (46) 1 = relapse (56) 2 = competing event (75)

